

HDI Predictor Documentation

Project Executable Files

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Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor - Human Development Index Classification System
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Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt	Yes
4	Database schema / seed files	NA (flat CSV dataset, no DB)
5	Environment configuration file (.env.example)	NA (no external API keys required)
6	Deployed application URL (if hosted)	Optional — runs locally via Docker
7	APK / Executable binary	NA
8	Dockerfile / Containerization config	Yes
9	Test files and test results	Yes
10	Demo video or walkthrough	Optional

Step 2: File / Folder Structure

```
hdi-predictor/
├── data/
│   └── hdi_country_indicators.csv
├── notebooks/
│   └── model_training_and_evaluation.ipynb
├── model/
│   ├── model.pkl
│   └── scaler.pkl
├── app/
│   ├── app.py
│   ├── preprocess.py
│   └── templates/
│       ├── index.html
│       └── result.html
├── tests/
│   └── test_predict.py
├── Dockerfile
├── requirements.txt
└── README.md
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Runs locally at <code>http://127.0.0.1:5000</code> (or 5050 if configured) after Docker/Flask launch
Login Credentials (Demo)	Not applicable — no authentication required for this evaluation tool
Platform / Hosting Provider	Local Docker container (portable to Render/Heroku/AWS if hosted publicly)
Repository Link	<code>https://github.com/saivenkata-A7/hdi-predictor</code> (placeholder — update with actual repository URL)
Demo Video Link	(to be attached with submission)

Step 4: Run Instructions

Option A — Docker (recommended):

```
docker build -t hdi-predictor .
docker run -p 5000:5000 hdi-predictor
```

Then open `http://127.0.0.1:5000` in a browser.

Option B — Local Python environment:

```
python -m venv venv
venv\Scripts\activate      # Windows
source venv/bin/activate   # macOS/Linux
pip install -r requirements.txt
python app/app.py
```

Then open `http://127.0.0.1:5000` and enter the four HDI indicators to get a prediction.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Model trained on a static, historical HDI dataset; predictions for extreme out-of-range indicator values may be less reliable	Documented input range guidance shown on the form
2	Flask development server is single-process by default	For production use, run behind Gunicorn inside the same Docker image
3	No persistent database for storing prediction history	Out of scope for the current evaluation tool; noted as a future enhancement